Rayan Daod Nathoo

ML Research Engineer

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French citizen, based in Switzerland

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I am naturally drawn to **impactful and technically challenging problems**, with a soft spot for **applied research**. At my best when I need to innovate under **hard constraints**, I move comfortably between **research and production**, and across fields, which makes me good at translating ideas from one domain into another. I learn fast, am hard to discourage, and I share what I learn as I go. So far I worked at **Bose, Logitech, EPFL, and Atinary**, where I designed, developed, and productized novel algorithms. With experience in **ML Research & Engineering**, I am a team worker eager to learn and looking forward to the next challenge!

TECHNICAL SKILLS

Python | TensorFlow/PyTorch/JAX | Bayesian Machine Learning | Audio ML | Agentic AI | Model Compression | Data Science | Data Engineering | Containerization | Software Engineering | System Design | Git

WORK EXPERIENCE

ML Research Engineer
Atinary

Startup developing AI, software, laboratories, to accelerate R&D in science

Mar, 2026 - Present | Lausanne, Switzerland

- Developing novel **Bayesian Optimization** methods to accelerate hypothesis search across chemical and materials science spaces (surrogate modeling, active learning, transfer learning)
- Constantly **reviewing the literature** to translate product requirements into research prototypes
- Maintaining and improving our core production algorithms and benchmarking pipeline

Machine Learning Engineer
Atinary

May, 2025 - Mar, 2026 | Lausanne, Switzerland

- Maintained and improved a highly available **inference** pipeline serving personalized models with evolving dependencies - PyData Talk
- Developed a scalable **training** pipeline accounting for latency, cost, reproducibility, long-term maintainability (AWS Batch, S3, Celery, MLflow)
- Migrated the ML codebase from multiple repositories to a **monorepo** and designed the tooling for developer experience - PyCon Talk

Junior Software Engineer
Atinary

Dec, 2023 - May, 2025 | Lausanne, Switzerland

- Owned and maintained the SDLabs **backend** (Microservices, OpenAPI, SQLAlchemy), and implemented user-requested features
- Implemented major **software architecture** changes (e.g., merging two REST API services), profiled, analyzed, and optimized database queries
- Drastically improved the team's **developer experience** (e.g., Docker builds, local environment)

Machine Learning Researcher
Bose

Feb, 2023 - Aug, 2023 | Internship - Boston, MA, USA (remote)

- Developed a novel **Knowledge Distillation (KD)** method for Speech Enhancement particularly good in adverse conditions: ICASSP publication
- Explored various **model compression** techniques (quantization, pruning) and implemented a general purpose KD framework
- Designed & improved product-level Deep Learning models (Speech Enhancement, Source Separation), using U-Net-like, DCCRN, TCN models

Contact
Mikolaj Kegler (Audio ML Scientist)

Research Assistant
Multimedia Signal Processing Group (EPFL)

May, 2022 - Aug, 2022 | Fixed-term contract - Lausanne, Switzerland

- Evaluated existing **Semantic Segmentation** (CNN) and **Super-Resolution** (GAN) methods with latency and hardware constraints
- Deployed models in a real-time TV studio pipeline (Norway, UK, ...) - Code, Publication, Reports - project validated by the EU Commission

Machine Learning Researcher
Logitech

Aug, 2021 - Mar, 2022 | Internship - Lausanne, Switzerland

- Developed real-time Deep Learning Target Source Separation models: 1/ footstep extractor for gaming 2/ sound type manipulator
- Developed an end-to-end pipeline: data augmentation, modeling (CNN, RNN), evaluation
- Demonstrated prototypes to the entire R&D department, UX Designers, Product Managers. Wrote and defended my Master's Thesis

SELECTED PUBLICATION

Two-Step Knowledge Distillation for Tiny Speech Enhancement

Published at ICASSP 2024

September 2023

EDUCATION

BSc & MSc in Computer and Communication Sciences
Swiss Federal Institute of Technology Lausanne (EPFL)
Jan, 2015 - Jan, 2022 | Lausanne, Switzerland

- Machine Learning, Deep Learning, Image Processing, Audio & Speech Processing, Biomedical Signal Processing, Adv. Algorithms, Adv. Probabilities
- Master's Thesis: "Personalised Audio Enhancement" with Logitech's CTO office and the LCAV laboratory (EPFL)
- Semester Project at LCAV: Percussion Sample Classifier - Data Engineering, Feature Extraction, Time-Frequency Analysis, Random Forest, CNN
- More projects: Twitter Sentiment Classification, Multi-Channel Noise Reduction, Direction Of Arrival Estimation, JUCE VST Plugin (C++ app)
- Activities: Master students representative, Event Manager -> Sponsoring Manager -> President of the Computer Science Student Association

CERTIFICATES

Coursera - TensorFlow Advanced Techniques
Custom Layers, Custom Loss Functions, Custom and Distributed Training, Advanced Computer Vision, Generative Deep Learning (4 courses – 80 hours)

Coursera - TensorFlow Developer Specialization
Convolutional Neural Networks, Natural Language Processing, Sequences, Time Series prediction (4 courses – 80 hours)

Coursera - Deep Learning Specialization
Neural Networks, Deep Learning, Hyper-parameter tuning, Regularization, Optimization, Convolutional Neural Networks, Sequence models (5 courses – 120 hours)

LANGUAGES

French
Native

English
Full Professional Proficiency (C2)

Spanish
Elementary Proficiency (A2)

HOBBIES

AI/tech advances · Music Production · Self-improvement · Traveling